

Structural Differences Between Adult and Non-Adult Films: A Data-Driven Analysis of Ratings, Genre Complexity, and Box Office Performance (2015–2020)

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1. Introduction

Prior research has shown that online movie ratings are not fully independent, as users' evaluations can be influenced by existing ratings and social dynamics. In particular, information cascades may lead users to align their opinions with prevailing trends, meaning that ratings may not always reflect true quality (Lee et al., 2015). However, existing studies primarily treat rating bias as a general phenomenon, with limited attention to whether such effects vary systematically across different content categories. This gap is particularly important, as films of different types may occupy distinct positions within the market and attract different audience groups. In particular, it remains unclear whether restricted or adult-oriented content may exhibit distinct rating patterns and market characteristics due to differences in audience composition or social perception.

This study aims to examine whether films with different content classifications occupy structurally different positions within the same platform environment. Specifically, we focus on adult and non-adult films produced in the United States and Canada between 2015 and 2020, using IMDb and related datasets as a unified analytical framework. In this study, “adult films” do not refer strictly to explicit or pornographic content, but rather to films with age restrictions (e.g., R-rated or 18+). Due to variations in classification systems across countries and platforms, we apply a standardized categorization during data processing, which will be described in detail in later sections.

Although platforms such as IMDb are often considered neutral systems where all films are evaluated under the same framework, films may not function equally within the market. In particular, adult films often face distribution constraints and limited audience accessibility. These structural differences may in turn affect their ratings, genre composition, and box office performance. Therefore, rather than evaluating artistic quality, this study focuses on whether films with different labels exhibit distinct behavioral patterns within the same data structure.

This study addresses the following research questions:

- RQ1: Do adult films exhibit systematically different IMDb and Rotten Tomatoes rating distributions compared to non-adult films?
- RQ2: Are adult films more likely to have concentrated or repetitive genre combinations compared to non-adult films?
- RQ3: Does adult content significantly affect international box office performance after controlling for IMDb ratings?

Understanding these questions is important because large entertainment datasets are often treated as objective representations of audience preferences. If systematic differences exist, this would suggest that platform data reflects underlying industry structures and participation patterns, rather than purely audience evaluations. These findings may contribute to a better understanding of how film data represents broader market dynamics.

2. Data

2.1 Datasets and Sources

Dataset	Source	Description
IMDb Basics	https://datasets.imdbws.com/	The IMDb dataset provides detailed metadata for films, including title identifiers, release year, genres, and IMDb user ratings. This dataset serves as the core foundation of our analysis because it provides the main film records and rating information needed to compare adult and non-adult films.
IMDb Ratings		
IMDb AKA's		
Streaming Dataset	https://www.kaggle.com/datasets/ruchi798/movies-on-netflix-prime-video-hulu-and-disney	This dataset contains movie information from several streaming platforms, including title, release year, age classification, and Rotten Tomatoes ratings. It is used to obtain the age rating that allows us to classify films as adult (18+) or non-adult, and to include Rotten Tomatoes ratings for comparison with IMDb ratings.
Box Office Dataset	https://www.kaggle.com/datasets/harios/box-office-data-1984-to-2024-from-boxofficemojo	This dataset provides box office revenue data for films, including movie title, release year, and total gross revenue. It allows us to analyze the financial performance of films and examine whether adult classification is associated with differences in box office outcomes.

These datasets were chosen because they offer complementary information necessary to address all three research questions in a unified analysis. The IMDb datasets supply essential movie metadata and audience ratings, while the streaming dataset contributes content classification (such as adult versus non-adult) along with critic ratings. Box office data provides measures of financial performance, and the AKA's dataset enables geographic filtering, specifically for films associated with the United States and Canada. By integrating these sources, we created a comprehensive movie-level dataset that supports the examination of ratings, genre structure, and revenue.

2.2 Data Cleaning

Each dataset was cleaned separately to ensure consistency in variables, formats, and filtering conditions prior to merging. First, only relevant columns were loaded from each dataset to reduce unnecessary processing. Filtering was then applied: the IMDb Basics dataset was restricted to movies only (titleType = "movie"), while the IMDb Basics, Streaming, and Box Office datasets were further limited to films released between 2015 and 2020. The IMDb AKA's dataset was filtered to include only records from the United States and Canada. Missing values were handled by replacing placeholder entries such as "\N" with NA, and rows with missing values in key variables were removed. Data types were standardized by converting year fields to numeric and removing invalid entries, transforming rating variables (IMDb and Rotten Tomatoes) into numeric format, and converting box office gross values from string format (e.g., "\$123,456") into numeric values. Additionally, column names were standardized across datasets to ensure consistency, particularly for shared variables like title and year. Duplicate records were also addressed: duplicates in the IMDb AKA's dataset were removed based on titleId, while duplicates in the Streaming and Box Office datasets were eliminated using title-year combinations. Finally, each cleaned dataset was saved as a separate CSV file for use in subsequent analysis.

2.3 Data Integration

After cleaning each dataset individually, the datasets were integrated to construct a unified movie-level dataset for analysis. IMDb Basics was used as the primary reference table, with movieID serving as the core identifier for merging related information. IMDb Ratings data was directly joined to the IMDb Basics dataset using movieID, ensuring accurate linkage of rating information. For datasets that did not share a common identifier, including the streaming and box office datasets, records were matched based on a combination of movie title and release year. While this approach enables broader data coverage, it may introduce potential matching inconsistencies, which are discussed in the limitations section. In addition, the IMDb AKAs dataset was used to identify geographic information. Region data was aggregated and used to filter films associated with the United States and Canada, ensuring consistency with the study's scope. Following integration, the resulting dataset contains all key attributes required for analysis, including ratings (IMDb and Rotten Tomatoes), genre information, content classification (adult vs. non-adult), and box office performance. This unified structure enables consistent analysis across all research questions at the movie level.

2.4 Exploratory Data Analysis

During exploratory data analysis, several important patterns and data issues were identified. First, missing data was unevenly distributed across variables: IMDb ratings were largely complete, whereas Rotten Tomatoes scores and box office data were more limited. This meant that applying stricter filters, such as requiring revenue data for certain analyses, reduced the sample size and introduced bias toward more popular or well-documented films. Second, outliers were observed in both ratings and revenue. Boxplots revealed several low-end outliers in IMDb and Rotten Tomatoes scores, while box office revenue was highly right-skewed, with a small number of blockbuster films dominating the distribution. These outliers were retained because they reflect real-world extremes, though visualization techniques such as transparency were used to reduce their visual impact. Third, rating distributions showed that both IMDb and Rotten Tomatoes scores were generally concentrated in the 50–70 range. Adult films tended to have a tighter distribution, particularly around 60–70, while non-adult films exhibited a wider spread, including more extreme low and high ratings. Finally, genre analysis revealed that most films are associated with multiple genres, confirming the need to treat genres as multi-valued attributes. Non-adult films displayed greater diversity in genre combinations, whereas adult films were concentrated in fewer, more repetitive genre patterns, supporting further analysis of genre complexity.

2.5 Limitations and Potential Biases

The analysis is subject to several limitations and potential sources of bias. First, dataset coverage is skewed toward films available on major platforms or those with recorded box office revenue, which likely overrepresents more popular or commercially successful productions. Second, the process of merging datasets using title and year may introduce matching errors due to inconsistent naming conventions or duplicate titles. Third, the “isAdult” variable is based on age ratings (e.g., “18+”), which may not perfectly capture actual adult content, leading to possible misclassification. Fourth, filtering out records with missing values creates bias by favoring films with more complete information, often excluding smaller or less-documented productions. Fifth, geographic filtering using AKAs regions (US/CA) serves only as an approximation and does not necessarily reflect the true production origin of films. Finally, box office revenue is influenced by many external factors not captured in the dataset, such as marketing, distribution scale, and franchise effects, so any observed relationships should be interpreted as associations rather than causal conclusions.

3. Methodology

RQ1: Rating Distribution Analysis

We classified films into adult and non-adult groups based on the “isAdult” variable, and compared their IMDb and Rotten Tomatoes rating using histograms and boxplots. The histograms allowed us to examine the overall distribution of ratings, including where ratings were concentrated and whether the two groups differed in spread or skewness. The boxplots were used to compare median, interquartile range, and outliers more clearly. These methods were chosen because RQ1 focuses on whether adult and non-adult films show different rating distributions, not just different average scores. Overall, we analyzed the data by combining visual comparison and descriptive statistics to evaluate differences in central tendency, variability, and distribution shape across the two rating sources.

RQ2: Genre Complexity Analysis

For each film, we split the genre string into individual genre labels, sort them alphabetically, and recombine them into a standardized genre combination string (e.g., “Crime|Drama|War”). This ensures that combinations such as “War, Crime” and “Crime, War” are treated as identical. Films are then divided into adult and non-adult groups based on the “isAdult” variable.

Genre complexity is analyzed in two ways. First, we compute the frequency of each full genre combination within each group, rank them, and identify the top 10 most common combinations. These are visualized using bar charts to assess whether adult films rely on a smaller set of repeated genre combinations. Second, we count the total appearances of individual genre labels across films in each group and use genre frequency bar charts and composition charts to compare broader genre patterns.

To further measure concentration, we calculate Top10Share for each group:

$$Top10Share = \frac{\text{Number of films in top 10 combinations}}{\text{Total number of films in group}}.$$

After splitting combinations into individual genres, we also compute Top5Share and Top10Share for single genres to assess how concentrated each group is in a small set of core genres. Overall, the analysis uses frequency analysis, ranking, bar-chart visualization, and concentration measures to evaluate whether adult films exhibit a more concentrated and repetitive genre structure compared to non-adult films.

RQ3: Revenue Structure

We modeled international box office revenue as the outcome variable, used isAdult as the key explanatory variable, and included IMDb rating as a control variable. Because box office revenue was highly skewed, we applied a log transformation before analysis. We first estimated a multiple linear regression model to test whether adult status significantly affected revenue after controlling for IMDb ratings.

$$\log(BoxOffice_i) = \beta_0 + \beta_1 * isAdult_i + \beta_2 * IMDbRating_i + \varepsilon_i$$

We then added an interaction term between isAdult and IMDb rating to examine whether the effect of ratings on revenue differed between adult and non-adult films.

$$\log_{10}(BoxOffice_i) = \beta_0 + \beta_1 * isAdult_i + \beta_2 * IMDbRating_i + \beta_3 * isAdult_i * IMDbRating_i + \varepsilon_i.$$

We evaluated the statistical significance, direction, and magnitude of the coefficients, and used scatterplots with fitted regression lines to help interpret the relationship visually. This approach was chosen because RQ3 asks whether adult status affects revenue while controlling for another important variable, making regression the most appropriate method.

4. Database Implementation

4.1 Oracle Schema Design and Key Decisions

The Oracle schema follows a normalized relational design consisting of three tables: Movies, Ratings, and BoxOffice. The schema was constructed by integrating five cleaned datasets into a unified structure.

The original datasets contained overlapping attributes such as titles, years, and identifiers, which would lead to redundancy and potential inconsistency if stored separately. To address this, the data was consolidated using a single movieID as the primary key, and attributes were grouped based on their relationships. The Movies table stores core information such as title, year, genres, and age classification, while Ratings and BoxOffice store additional attributes linked through one-to-one relationships using the “movieID” foreign key.

This structure ensures data integrity and avoids duplication, although it requires join operations to combine data across tables. One limitation of this design is that the “genres” field is stored as a single string, which makes genre-based analysis less flexible and requires additional parsing.

4.2 MongoDB Document Design and Key Decisions

In MongoDB, the schema is designed into a single movies collection where each document represents one movie and contains all related information. Instead of separating data into multiple tables, ratings and box office information are stored as embedded subdocuments within each movie document.

This design reduces the need for joins and is better suited for the project’s query patterns, which frequently analyze movies together with their ratings and revenue. The ratings and boxOffice fields are embedded because they have a one-to-one relationship with each movie and are always accessed together. Using references would require join operations, which would reduce the efficiency advantages of MongoDB’s document model.

Additionally, “genres” is stored as an array of strings rather than a comma-separated value, allowing more flexible and efficient queries. The “isAdult” field is stored as a boolean derived from the original age classification to simplify filtering.

4.3 Comparison of Database Approaches

The MongoDB approach felt more natural for this project because the data is inherently movie-centric, and most queries require accessing all attributes of a movie at once. Storing all related data in a single document simplifies queries and improves read performance. In contrast, the Oracle design is more structured and avoids redundancy, but requires joins to answer the same analytical questions, making it less convenient for this use case.

5. Database Comparison and Reflection

In this project, I would choose MongoDB because the main unit of analysis is the movie, and all relevant attributes (ratings, genres, and box office) are naturally associated with a single movie. In our MongoDB design, each movie is stored as one document containing all related information, which allows queries to access the required data without joins. In contrast, Oracle requires joining multiple tables (Movies, Ratings, and BoxOffice) for almost every research question. Since our analysis is read-heavy and repeatedly combines the same attributes, MongoDB provides a more direct and efficient way to retrieve and aggregate data.

However, Oracle would become more suitable if the project involved more complex relationships instead of a movie-centric structure. For example, if each movie had multiple rating records over time (a one-to-many relationship), if box office data were stored at a more granular level (such as per country or per week), or if the analysis frequently required linking multiple entities (such as actors, directors, or production companies), then the relational model would be more appropriate. In these cases, normalization and structured joins in Oracle would help maintain data consistency and support more complex analytical queries.

When comparing query implementation across research questions, the relative advantages of each system become more evident. For RQ1 (rating distribution), MongoDB made the query easier. Its aggregation pipeline allows summary statistics (mean, minimum, maximum, and standard deviation) as well as full distributions to be computed directly within the database. In contrast, the SQL version mainly retrieves row-level data, requiring additional processing in Python to produce the same results. As a result, MongoDB is more efficient and better suited for this type of distribution analysis.

For RQ2 (genre complexity), MongoDB provided a significantly clearer and simpler approach. Because genres are stored as arrays, MongoDB can directly use the `$unwind` operation to separate and count genres, making aggregation straightforward and intuitive. In contrast, Oracle stores genres as comma-separated strings, which requires string parsing and additional transformation steps before aggregation. This increases both complexity and implementation effort in SQL.

For RQ3 (revenue structure with control for IMDb rating), the two systems are more comparable. MongoDB still has an advantage in terms of data access, as all relevant attributes (`isAdult`, `ratings.imdb`, and `boxOffice.gross`) are stored within a single document, allowing filtering and aggregation to be performed in one pipeline without joins. However, Oracle provides a clearer structure for expressing analytical logic using SQL, particularly when modeling relationships between variables. Therefore, neither system has a strong advantage for this question, although MongoDB slightly simplifies data retrieval.

Overall, some queries were significantly easier in MongoDB. The most notable example is the genre analysis in RQ2, where MongoDB's array-based design allows direct aggregation using `$unwind` and `$group`. In contrast, the SQL implementation requires splitting comma-separated strings, which adds complexity and reduces readability. Similarly, the distribution analysis in RQ1 is more straightforward in MongoDB, since the aggregation pipeline can directly compute both summary statistics and distributions, while SQL requires additional processing outside the database.

6. Results

RQ1: Rating Distribution Analysis

Between 2015 and 2020, do adult films exhibit systematically different IMDb and Rotten Tomatoes rating distributions compared to non-adult films produced in the United States and Canada?

IMDb vs Rotten Tomatoes Rating Distribution: Adult vs Non-Adult Films (2015–2020)

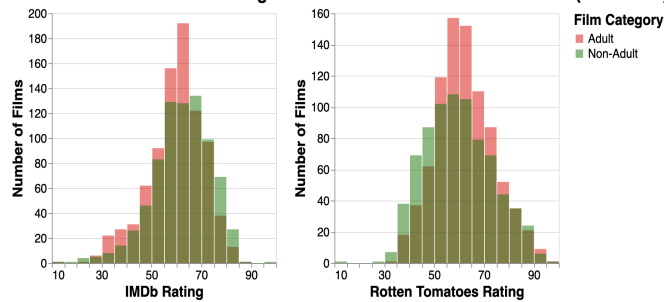


Figure 1

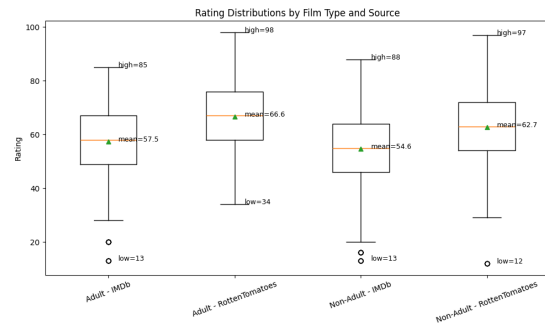


Figure 2

Visualization Interpretation

The histogram shows the full distribution of ratings for both groups. For IMDb ratings, both adult and non-adult films are concentrated between 50 and 70, but adult films appear more tightly clustered around the mid-range (especially near 60). Non-adult films exhibit a wider spread, including more observations at both lower (<30) and higher (>75) rating ranges.

A similar pattern is observed for Rotten Tomatoes ratings. While both groups peak around 50-65, non-adult films again display greater variability, with more extreme values on both ends. Adult films, in contrast, are more concentrated and show fewer extreme highs and lows.

The boxplot further confirms these patterns. Adult films have slightly higher average ratings (IMDb ≈ 57.5 vs 54.6 ; Rotten Tomatoes ≈ 66.6 vs 62.7) and a narrower interquartile range, indicating more consistency. Non-adult films show larger spread and more outliers, suggesting greater variability in audience and critic reception.

Statistical Comparison and Distribution Shape

From both visualizations, the statistical comparison and distribution shapes show clear differences between adult and non-adult films. Adult films tend to have slightly higher mean ratings on both platforms, along with lower variability, indicating more concentrated distributions and fewer extreme values. In contrast, non-adult films exhibit greater dispersion, with a wider spread of ratings and more outliers, reflecting a broader range of performance that includes both very low and very high scores. Overall, while both groups are roughly centered in the mid-range of ratings, non-adult films display a more dispersed distribution, whereas adult films are more tightly clustered.

Conclusion and Hypothesis Evaluation

Our initial hypothesis suggested that adult films would generally receive lower ratings compared to non-adult films. However, the results do not support this hypothesis. Instead, adult films have slightly higher average ratings and more consistent performance, while non-adult films demonstrate greater variability in quality. This suggests that although adult films are fewer in number, they may represent a more niche but consistently rated category, whereas non-adult films cover a broader spectrum of quality and audience reception.

RQ2: Genre Complexity Analysis

Between 2015 and 2020, are adult films in the U.S. and Canada more likely to combine multiple genre tags compared to non-adult films?

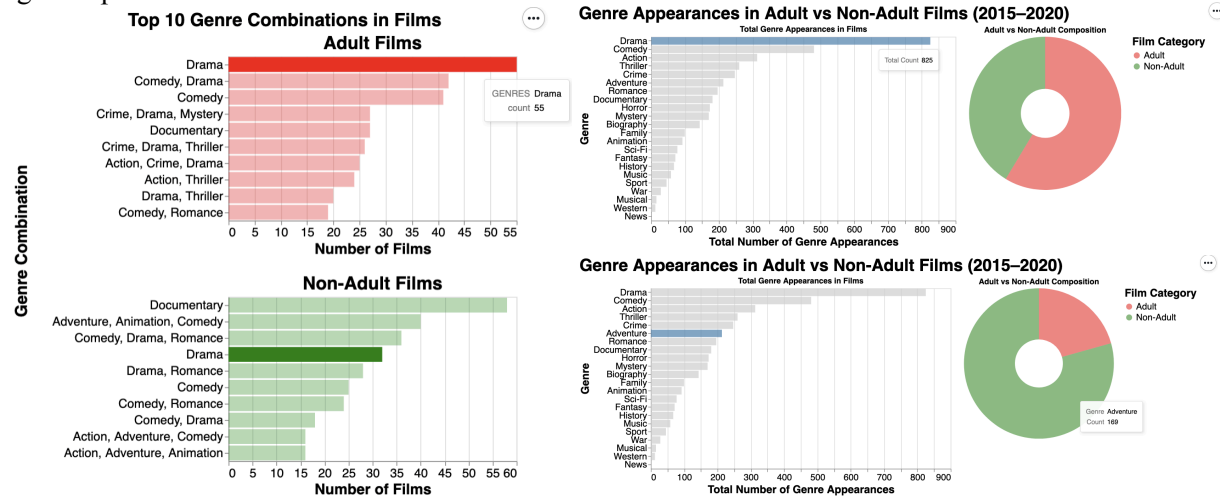


Figure 3

Figure 4

We analyzed the data from three angles: the top 10 full genre combinations (Figure 3), the distribution of individual genre appearances (Figure 4), and the TopShare measures based on both full combinations and single genres.

Figure 3 presents the most frequent genre combinations for each group. Adult films are dominated by drama- and thriller-related combinations. In contrast, non-adult films span a wider range of family-, adventure-, and documentary-oriented patterns. However, the Top10Share based on full combinations is 36.2% for adult films and 39.1% for non-adult films, indicating that adult films are not more concentrated at the combination level.

Figure 4 shifts the focus to individual genres. The bar charts show the total frequency of each genre, and the donut charts show the adult versus non-adult composition for selected genres. Drama is the most common genre overall, followed by Comedy, Action, Thriller, and Crime. Adult films are more strongly associated with Drama, Thriller, and Crime, while non-adult films show greater representation in genres like Adventure and Romance. For example, Thriller and Crime have noticeably larger adult shares, whereas Adventure is much more heavily represented in non-adult films. This suggests that adult films are more concentrated in darker and more intense genre types, while non-adult films are spread across a broader range of mainstream genres.

TopShare measures for individual genres reinforce this pattern. The Top5Share is 64.6% for adult films versus 58.4% for non-adult films, and the Top10Share is 88.2% versus 81.7%, respectively. This indicates that adult films are more concentrated in a smaller set of core genres when analyzed at the individual genre level. This shows that once genre combinations are broken down into single labels, adult films are more concentrated in a small set of core genres.

Overall, the results provide partial support for our initial hypothesis. If genre complexity is defined using full genre combinations, the hypothesis is not supported, since adult films are slightly less concentrated than non-adult films according to combination-based Top10Share. However, if genre complexity is interpreted in terms of single-genre composition, the hypothesis is supported: adult films are more concentrated in a smaller set of dominant genres, especially Drama, Thriller, and Crime, while non-adult films display broader genre diversity.

RQ3: Revenue Structure

In films produced by the United States or Canada between 2015 and 2020, does adult content significantly affect international box office performance after controlling for IMDb ratings?

To examine whether adult content affects international box office performance, a multiple linear regression model was estimated for films produced in the United States or Canada between 2015 and 2020. The dependent variable is the logarithm of international box office revenue, while the key independent variable is an indicator for adult content. IMDb rating is included as a control variable.

$$\text{Model: } \log_{10}(\text{BoxOffice}_i) = \beta_0 + \beta_1 * \text{isAdult}_i + \beta_2 * \text{IMDbRating}_i + \varepsilon_i$$

The coefficient on isAdult is negative and statistically significant ($p < 0.001$). This indicates that adult films earn less revenue than non-adult films after controlling for IMDb ratings. IMDb ratings are positively and significantly associated with revenue, suggesting that higher-rated films generally perform better at the box office.

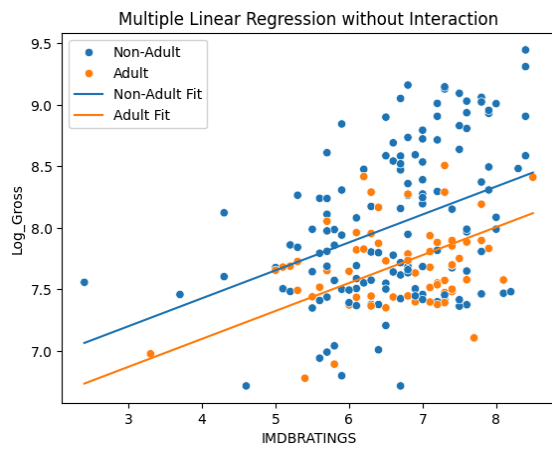


Figure 5

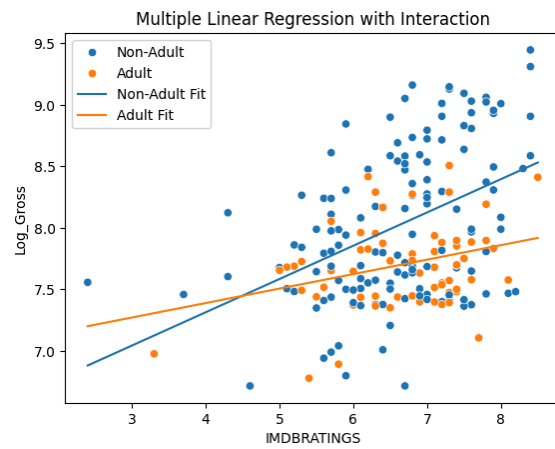


Figure 6

To further investigate whether the effect of ratings differs by content type, an interaction term between isAdult and IMDb ratings was introduced:

$$\log_{10}(\text{BoxOffice}_i) = \beta_0 + \beta_1 * \text{isAdult}_i + \beta_2 * \text{IMDbRating}_i + \beta_3 * \text{isAdult}_i * \text{IMDbRating}_i + \varepsilon_i$$

After including an interaction between isAdult and IMDb ratings, the direct effect of adult content is no longer statistically significant ($p = 0.207$). However, the interaction term is negative (-0.1529) and marginally significant ($p = 0.060$), suggesting that the positive effect of IMDb ratings on revenue is weaker for adult films compared to non-adult films. In other words, while higher ratings generally increase revenue, this benefit is smaller for adult films.

Figures 5 and 6 present scatter plots of IMDb ratings against log-transformed revenue for the models without and with interaction, respectively. In Figure 5, the fitted regression lines for adult and non-adult films are parallel, indicating that IMDb ratings have a similar effect on revenue across both groups, with adult films consistently earning less. In Figure 6, the fitted lines are no longer parallel. While both groups exhibit a positive relationship between ratings and revenue, the slope for adult films is flatter, indicating that increases in IMDb ratings lead to smaller gains in revenue compared to non-adult films. This visual difference supports the presence of an interaction effect between adult content and ratings.

Overall, the results partially support the initial hypothesis. While the baseline model suggests that adult films earn lower revenue, this effect is not robust after accounting for interaction effects. Instead, the findings indicate that adult content moderates the relationship between ratings and revenue, with higher ratings having a weaker impact on revenue for adult films.

7. Limitations and Future Work

7.1 Limitations

This study has several limitations related to data, methodology, and validity. These limitations represent potential threats to validity, as they may influence the observed relationships but cannot be directly verified or fully controlled in this study. First, the datasets used in this project do not fully represent the entire film industry. The data primarily includes films that are available on major platforms or have recorded ratings and box office information. As a result, the sample is likely biased toward more popular or commercially visible films, while smaller or less-documented productions are underrepresented. Second, the integration of multiple datasets may introduce matching errors. Since some datasets were linked using title and year rather than a unique identifier, inconsistencies in naming conventions or duplicate titles may lead to incorrect matches. This could affect the accuracy of variables such as revenue or content classification. Third, the classification of adult films is based on age ratings (e.g., “18+”), which may not perfectly capture the true nature of adult-oriented content. Some films may be misclassified due to differences in rating systems across platforms, introducing measurement error into the key independent variable. Fourth, missing data may introduce bias. Observations with missing values in key variables were removed, which reduces sample size and may systematically exclude certain types of films, particularly those with limited exposure or incomplete records. Fifth, the regression analysis in RQ3 does not account for other important factors that influence box office performance, such as marketing budget, franchise effects, star power, or distribution scale. As a result, the estimated relationships should not be interpreted as causal. Finally, the study focuses only on films produced in the United States and Canada between 2015 and 2020. This limited scope may reduce the generalizability of the findings to other regions, time periods, or film markets. In addition, the primary data sources used in this study, such as IMDb and Rotten Tomatoes, are North America–focused platforms. As a result, the dataset largely reflects films that are distributed and evaluated within the North American market, making it difficult to capture audience behavior and market dynamics in other regions, such as Asia or Europe.

7.2 Future Work

If more time and resources were available, several extensions could improve and deepen this study. First, future work could incorporate additional variables, such as production budget, number of theaters, or marketing intensity, to better explain variation in box office performance and improve the regression model. Second, a more robust data integration process could be implemented using unique identifiers across datasets, reducing potential matching errors and improving data reliability. Third, alternative definitions of adult content could be explored, such as using multiple indicators or external classification sources, to reduce measurement error and improve the accuracy of content categorization. Fourth, more advanced analytical methods could be applied. For example, causal inference techniques or panel data models could be used to better isolate the effect of adult content on ratings and revenue over time. Finally, future research could expand the scope of the dataset to include films from additional countries or longer time periods. This would allow for cross-market comparisons and provide a more comprehensive understanding of how content classification influences film performance globally.

Beyond methodological extensions, this study also raises several new questions for future research. For example, why do adult films exhibit more consistent rating patterns despite lower market accessibility? Additionally, how do platform-specific audiences influence rating behavior across different regions? These questions were beyond the scope of this study but warrant further investigation.

8. References

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9. AI Declaration

We do not use any AI tools in this assignment.